

Norms

Vector ki 'size' measure karne ke alag-alag tarike

01 Opening Hook

STORY TIME

Socho tumhe office jaana hai. Ek raasta hai 'seedhi line mein diagonal chalna' (jaise bird flies), doosra hai 'sirf roads follow karna (grid pattern mein)'. Dono raaste 'distance' measure karte hain, lekin differently. Vector ki 'size' measure karne ke bhi aise multiple tarike hote hain - inhe kehte hain ****Norms****.

02 Why Does This Concept Exist?

THE PROBLEM

AI models mein humein baar baar vectors ki 'size' ya 'magnitude' jaanni hoti hai - kitna bada error hai, weights kitne 'bade' ho gaye hain, ya do points kitne door hain. Norm invent hua taaki hum in sawalon ka precise, mathematical jawab de sakein, aur different situations ke liye different tarike se measure kar sakein.

03 Explain Like I'm 10

SIMPLE WORDS

Socho tum ek city mein ho jahan roads sirf grid pattern mein hain (jaise Manhattan). Agar tumhe A se B jaana hai, tum diagonal nahi ja sakte - sirf horizontal aur vertical. Yeh distance 'Manhattan distance' (L1 norm) kehlaata hai.

Agar tum khule maidan mein ho aur seedha (diagonal) ja sakte ho, woh 'straight line distance' (L2 norm, jo humara normal 'ruler se measure karna' hai) kehlaata hai.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Norm ek function hai jo kisi vector ko lekar ek single non-negative number deta hai jo uski 'size' represent karta hai. Alag-alag norms (L1, L2, etc.) size ko differently define karte hain - kuch outliers ko zyada penalize karte hain, kuch kam.

05 Ab Thoda Maths

FORMULA TIME

General p-norm:

$\|v\|_p = (|v_1|^p + |v_2|^p + \dots + |v_n|^p)^{1/p}$. $p=1$ par yeh L1 norm hai, $p=2$ par L2 norm hai.

Common Norms:

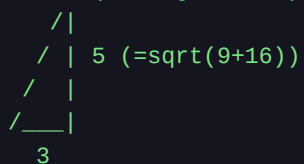
L1 norm = sum of absolute values. L2 norm = $\sqrt{\text{sum of squares}}$ - yeh 'normal' Euclidean distance hai.

06 Visualize Everything

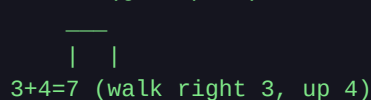
PICTURE IT

Vector $v = [3, 4]$

L2 Norm (straight line):


5 (=sqrt(9+16))

L1 Norm (grid path):


3+4=7 (walk right 3, up 4)

07 AI Connection

WHERE AI USES THIS

Loss Functions: MSE loss L2 norm use karta hai (squared errors), jabki MAE loss L1 norm use karta hai (absolute errors) - dono model errors ko differently penalize karte hain.

Regularization: L1 aur L2 regularization (weight decay) model weights ki norm ko control karte hain overfitting rokne ke liye.

08 Real Industry Example

REAL COMPANIES

All ML companies: Har training loop mein loss function ke andar norms (L1/L2) directly use hote hain error measure karne ke liye.

OpenAI/Anthropic: Gradient clipping ke dauraan gradient ka L2 norm check kiya jaata hai taaki training stable rahe (agar norm bahut bada hai, use scale down karte hain).

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch mein: ``torch.norm(v)`` (default L2), ``torch.norm(v, p=1)`` (L1). Gradient clipping: ``torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)`` gradients ke L2 norm ko limit karta hai training stability ke liye.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Norm sirf 'positive distance' measure karta hai simple tarike se.

Sahi baat: Different norms (L1 vs L2) bahut alag properties rakhte hain - L1 sparse solutions promote karta hai, L2 smooth solutions; yeh choice model behavior ko significantly affect karti hai.

Galat soch: L2 norm hamesha best choice hai.

Sahi baat: L1 norm outliers ke against zyada robust hota hai aur sparsity induce karta hai (kai weights ko exactly zero bana deta hai), jo feature selection ke liye useful hai.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Norm kya measure karta hai?

A: Vector ki 'size' ya magnitude, ek non-negative number ke roop mein.

[Intermediate] Q: L1 aur L2 norm mein practical farak kya hai?

A: L1 norm (Manhattan distance) outliers ke against zyada robust hai aur sparsity create karta hai; L2 norm (Euclidean distance) smooth hai lekin outliers se zyada affect hota hai.

[Advanced] Q: Gradient clipping mein norm ka role kya hai?

A: Training ke dauraan agar gradients bahut bade ho jaayein (exploding gradients), unka L2 norm calculate karke unhe ek maximum threshold tak scale down kiya jaata hai, jisse training stable rehti hai.

12 Exercises

PRACTICE TIME

[Easy] Vector [3,4] ka L2 norm nikalo.

Solution: $\sqrt{9+16} = 5$.

[Medium] Vector [3,4] ka L1 norm nikalo.

Solution: $|3|+|4| = 7$.

[Hard] Kyun L1 regularization sparse solutions (kai weights ko zero) produce karta hai jabki L2 nahi?

Solution: L1 norm ka gradient constant magnitude rakhta hai chahe weight chhota ho ya bada, isliye optimization weights ko exactly zero tak push kar sakta hai; L2 ka gradient weight ke proportional hota hai, isliye woh weights ko chhota toh karta hai lekin exactly zero tak nahi le jaata.

13 Golden Summary

Concept	Easy Meaning
Norm	Vector ki size measure karne ka function
L1 Norm	Sum of absolute values – Manhattan distance
L2 Norm	$\sqrt{\text{sum of squares}}$ – straight-line distance

Regularization	Norms weight size control karte hain (overfitting rokne ke liye)
Gradient Clipping	L2 norm se gradients ko stable rakhna

14 One-Line Memory Trick

“Norm batata hai vector 'kitna bada' hai – bas measure karne ka tarika alag ho sakta hai.”